

GUIDE: A Benchmark for Understanding and Assisting Users in Open-Ended GUI Tasks

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Abstract

Graphical User Interface (GUI) agents have the potential to assist users in interacting with complex software (e.g., PowerPoint, Photoshop). While prior research has primarily focused on automating user actions through clicks and keystrokes, this paradigm overlooks human intention, where users value the ability to explore, iterate, and refine their ideas while maintaining agency. To move beyond automation and toward collaboration, GUI agents must understand what users are doing and why. We introduce **GUIDE** (GUI User Intent Detection Evaluation), a benchmark that evaluates AI models on their ability to perceive user behavior, infer intent, and provide assistance in open-ended GUI tasks. GUIDE consists of 67.5 hours of screen recordings from 120 novice user demonstrations with think-aloud narrations, across 10 software. GUIDE defines three tasks—(i) Behavior State Detection, (ii) Intent Prediction, and (iii) Help Prediction that test a model’s ability to recognize behavior state, reason about goals, and decide when and how to help. Evaluations across eight state-of-the-art multi-modal models reveal that all models struggled, achieving only 44.6% and 55.0% accuracy on behavior state and help prediction. However, providing user context significantly improved the performance, raising help prediction by up to 50.2pp, highlighting the critical role of structured user understanding in effective assistance. Our dataset is available at <https://guide-bench.github.io>.

1. Introduction

Graphical User Interface (GUI) agents hold great promise for supporting users in complex workflows, in mobile [20, 29, 53], web [9, 16, 41, 50, 56], and software application tasks [10, 40, 51]. In creative and analytical tools such as Photoshop or PowerPoint, these agents can automate repetitive subtasks or provide guidance to help users achieve their goals more efficiently. Most existing GUI agents, both in academic research [14, 27, 28, 54] and in commer-

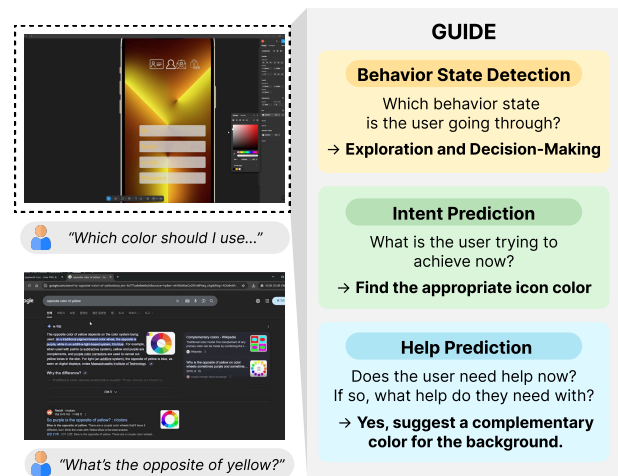


Figure 1. An example of the GUIDE benchmark, which jointly models three tasks: Behavior State Detection, Intent Prediction, and Help Prediction, to interpret what the user is doing, aiming to achieve, and whether and what they may need assistance with during open-ended software tasks.

cial services like Microsoft Office Copilot [32] or Figma Make [12], focus on full automation: given a goal, they either execute a sequence of clicks and keystrokes to complete the task or directly generate the desired output.

While this approach offers convenience, it overlooks how people actually work. In real-world open-ended workflows, success is not driven solely by efficiency—user satisfaction plays an equally critical role. Automated agents assume fixed goals, yet users frequently revise their intentions mid-task. For example, a user may reposition an element multiple times before reverting to the original—behavior an automated agent would treat as redundant, but which is essential for forming a preference. Rather than replacing user agency, effective assistance should accelerate exploration while keeping the user in control [22].

Recent work on proactive task assistance takes a more balanced approach [31, 45, 47, 48, 52]. Rather than auto-

Dataset	Domain	# Video	# Video Duration	Video Source	Primary Goal	Evaluation Focus		
						Behavior	Intent	Help
PsTuts [25]	1	-	71.4 h	Instructional Videos	Action Understanding			
VideoWebArena [19]	6	74	3.8 h	Human-Recorded Tutorials	Task Automation			
VideoGUI [27]	11	178	7.1 h	Instructional Videos	Task Automation		✓	
UI-Vision [33]	83	450	4.8 h	Experts Performing Tasks	Task Automation			
AssistGUI [14]	9	100	<8.3 h	Instructional Videos	Task Automation		✓	
WorldGUI [54]	10	611	<30.5 h	Instructional Videos	Task Automation		✓	
GUIDE (Ours)	10	120	67.5 h	Novice Users' Demonstrations	Behavior Understanding	✓	✓	✓

Table 1. **Comparison of GUIDE with existing GUI video understanding datasets.** GUIDE differs from existing benchmarks by (i) collecting screen recordings from novice users, (ii) capturing how they naturally behave in open-ended tasks with a focus on behavior understanding, and (iii) evaluating systems based on human user needs rather than task automation.

mate tasks for users, proactive assistants infer a user’s context and intent and deliver timely help. Studies in programming and productivity tools show higher efficiency and satisfaction when a system detects a need and intervenes at the right moment [6, 37, 38, 46]. Yet, the ability to model and track users’ evolving context remains underexplored in current multimodal systems that power GUI agents.

To achieve a truly human-assisting GUI agent, a key ability is to comprehend users’ cognitive context and intentions to provide appropriate support [17]. In real-world scenarios, users rarely articulate their goals or needs explicitly, making it natural for systems to rely primarily on visual cues from the screen. These user actions often carry semantic structure, such as hovering, undoing, or repeatedly opening menus, that signal intent. However, interpretation remains challenging: similar actions may stem from entirely different intents. For example, repeated undo actions might indicate confusion or deliberate refinement. Without deeper reasoning, assistance based solely on surface-level actions can lead to shallow or misaligned responses.

To address this challenge, we present **GUIDE** (GUI User Intent Detection Evaluation), a benchmark designed to evaluate multimodal LLMs (MLLMs) on their ability to understand and assist users in complex software workflows. GUIDE introduces a three-stage evaluation framework: (1) *Understanding* the user’s behavioral state to identify their current workflow phase; (2) *Reasoning* about their underlying intentions and goals; and (3) *Assisting* by delivering the appropriate form of help at the right moment.

We collected 67.5 hours of screen recordings from 120 human demonstrations across 10 widely used applications—including Photoshop, Figma, PowerPoint, Premiere Pro, and Excel—covering 40 open-ended tasks designed to elicit natural user behavior. Unlike prior work that primarily targets video understanding from expert-recorded instructional videos on closed-ended tasks [14, 25, 27, 33, 54], our focus is on novice users working on open-ended tasks, with the goal of building collaborative AI sys-

tems that assist users during exploration, trial-and-error, and learning. Observing novice workflows allows us to capture authentic moments of confusion, decision-making, and discovery, offering rich opportunities for AI to provide timely, context-aware support. Each session includes both screen recordings and think-aloud narrations that surface the user’s underlying intentions and cognitive states.

Building on this dataset, we define three-staged benchmark tasks: First, (i) **Behavior State Detection** evaluates whether a model can identify the user’s behavioral state, such as exploration or confusion, based solely on visual cues. To support this, we developed a taxonomy of nine user states reflecting diverse cognitive and behavioral phases in open-ended GUI workflows, grouped into four high-level categories: Planning, Execution, Problem-Solving, and Evaluation (Figure 3). This structure aligns with human cognition and interaction theories [4, 34], while introducing finer distinctions tailored to GUI-based task behavior. Next, (ii) **Intent Prediction** targets inference of the user’s immediate goal—what they are trying to accomplish in the given moment. The final task, (iii) **Help Prediction**, assesses whether a model can determine 1) whether the user needs assistance or not, and if so, 2) what type of help would be most appropriate, such as explaining a feature, suggesting an alternative, or addressing an error. By leveraging both visual screen recordings and accompanying think-aloud narrations, we automatically generated data for each task, which was subsequently verified through human review for accuracy and consistency.

Evaluation across eight state-of-the-art MLLMs reveals that while current models struggle to interpret user behavior and predict underlying intent and help needed—achieving only 44.6% accuracy on behavior state detection and 55.0% on help prediction, performance improves significantly when structured user context is provided. For example, supplying behavioral state and intent information boosted help prediction accuracy by up to 50.2 percent points for the lowest-performing model.

Our results suggest a promising path forward: providing different layers of human-grounded context, such as behavioral cues, inferred goals, and temporal history, can lead to more accurate assistance decisions. Our benchmark provides a foundation for training and evaluating the next generation of context-aware, collaborative GUI agents.

2. Related Work

2.1. Video Understanding meets GUI

Several benchmarks evaluate video understanding in the context of GUI and software workflows. Early work by Li et al. [25] collected Photoshop tutorial videos to understand screencast videos. More recent datasets span multiple applications and tasks. For example, AssistGUI [14] focuses on automating GUI tasks using an actor-critic agent, serving as a benchmark for task-oriented GUI automation. VideoWebArena [19] evaluates long-horizon multimodal agents on web browsing tasks, emphasizing extended video context and web UI interactions. VideoGUI [27] compiles high-quality instructional screen recordings and introduces a hierarchical model for mapping visual observations to GUI actions. UI-Vision [33] provides a fine-grained desktop UI video benchmark with dense annotations for perception and interaction. Lastly, WorldGUI [54] increases task diversity by allowing arbitrary initial interface states for each task, challenging agents to handle varied starting conditions.

These prior benchmarks primarily focus on close-ended tasks with predetermined goals, aiming to replicating expert demonstrations. In contrast, our work targets open-ended GUI workflows with novice users, emphasizing understanding of user intent and context rather than step-by-step replication of actions. This shift toward user-centric evaluation fills a gap not covered by existing GUI video datasets that evaluate task completion or action prediction.

2.2. Collaborative and Proactive Agents

While GUI agents that automate interface operations based on a given goal or instruction can be effective, this fully autonomous approach can conflict with the needs of users in creative or analytical environments, where retaining control and exploring alternatives are essential. To address this, recent research has shifted toward assistive GUI agents that collaborate with users by understanding context and offering timely support. Several works have explored inferring user goals and intent in both web [36] and software environments [3, 13, 55] to better align assistance with user needs. For example, Zhao et al. [55] introduce ProactiveVA, a visual analytics agent that monitors user interactions and leverages LLMs to detect when users may be stuck, providing context-sensitive suggestions or guidance.

Several recent works in the Human-Computer Interaction (HCI) community explore this shift toward collabora-

tion and contextual support. CowPilot [18] proposes a mixed-initiative framework that enables users to share control with an autonomous web navigation agent, improving efficiency while preserving agency. In programming, proactive assistants like Codellaborator [38] and NeedHelp [6] demonstrate how real-time intervention can aid users when well-timed. Studies on software applications [22] show users prefer AI agents that guide them rather than take over entirely, reinforcing the need for transparency and shared control. ProMemAssist [39] further highlights the benefits of modeling user cognition to deliver timely, non-intrusive support. These findings echo broader discussions on autonomy levels [11] and the importance of aligning agent behavior with human preferences [21, 26]. Our work builds on these insights, evaluating how well current multimodal models can perceive a user’s state and intentions in GUI workflow recordings and decide if and how to assist. By situating the evaluation in real user workflows, we aim to push GUI agents toward true user-aware collaboration.

3. GUIDE Benchmark

To develop a benchmark that focuses on understanding and assisting users, we collected demonstrations from novice users. Unlike existing datasets that focus primarily on expert demonstrations or polished instructional videos [14, 25, 27, 30, 33, 44, 54], our dataset captures the authentic challenges and exploratory behaviors that novices exhibit during task completion, serving a crucial role in building collaborative agents. Building on these demonstrations, we propose a suite of tasks designed to evaluate models’ capabilities to understand users and provide effective assistance.

3.1. Video Collection

We collected 120 demonstrations from novice users across 10 applications spanning five categories: Photo Editing (Photoshop, GIMP), Graphic Design (Figma, Canva), Presentation Design (PowerPoint, Google Slides), Video Editing (Premiere Pro, CapCut), and Data Analysis (Google Sheets, Microsoft Excel). For each application, we designed four open-ended tasks aimed at eliciting natural and diverse user behaviors and approaches (Table B2 in supp.).

We chose creative and analytical tools to surface exploratory workflows and variation in problem-solving strategies. Each task was completed by three different users to capture diverse strategies and behaviors. We ensured that each task was flexible enough, while still incorporating elements of challenge. Participants were asked to spend at least 20 minutes per task and meet a few minimal requirements (e.g., inserting a relevant image) to mark it as complete.

We recruited 54 novice users of software from Prolific and our institution. Participants were screened based on self-reported expertise to ensure novice-level familiarity with the features in the target application (Mean: 2.8, SD:

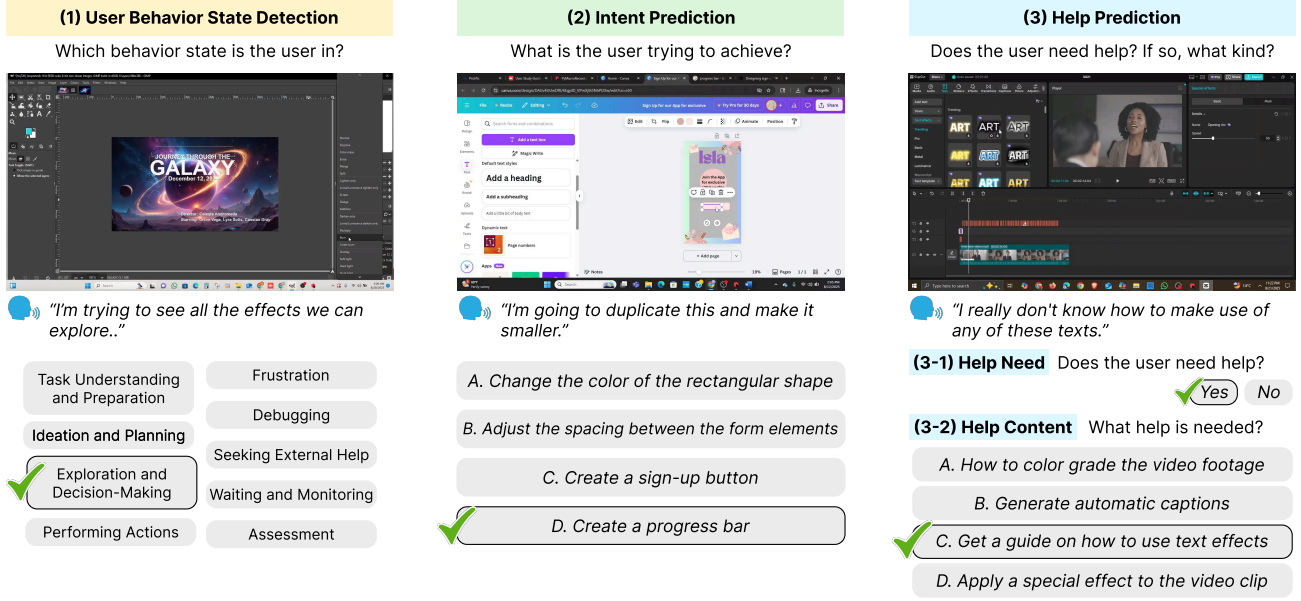


Figure 2. Overview of the three core tasks in the GUIDE benchmark. (1) **User Behavior State Detection** identifies the user’s current behavioral mode (e.g., *Exploration and Decision-Making*). (2) **Intent Prediction** infers what the user is trying to achieve (e.g., *Create a progress bar*). (3) **Help Prediction** determines whether the user needs assistance and, if so, what kind of help is relevant (e.g., *Get a guide on how to use text effects*). Together, these tasks enable a comprehensive understanding of user behavior and assistance needs in software GUI environments. We evaluate MLLMs on their ability to infer these solely from the visual input, without access to the demonstrator’s narration — a setting that closely reflects real-world use.

1.1, Range: 1–5). During the study, participants worked on the assigned task while recording their screen and keyboard/mouse input events. They were also asked to think aloud and record their voice as they carried out the task, verbalizing what they were doing and their thought process.

3.2. Benchmark Tasks

To evaluate a model’s ability to understand user context and deliver appropriate assistance, we design our benchmark as a unified three-stage framework: *Understanding* → *Reasoning* → *Assisting*. These stages progress from interpreting user behavior to inferring intentions and ultimately providing helpful assistance. Each task corresponds to a distinct level of cognitive inference required for a human-assisting GUI agent to effectively support users in open-ended software workflows.

To construct a dataset for task evaluation, we used the Human-AI collaborative method. We first transcribed the think-aloud narration using WhisperX [2], and used the narration as a main source for extracting initial annotations in addition to the video. We employed *Gemini-2.5-Pro* [15] to first create annotations needed for each task, which were then refined by human annotators. Note that we use narration only as an annotation source to capture users’ intentions and mental states. The benchmark evaluates vision-only understanding, testing whether models can infer these

states solely from visual cues, as in real-world settings with limited access to user speech.

3.2.1. User Behavior State Detection

Description. This task evaluates whether a model can interpret the user’s behavioral context directly from visual cues. Models are asked to classify a video segment into one of nine behavior states in our taxonomy (Figure 3), which spans the full range of cognitive and behavioral processes observed in creative and analytical workflows.

We developed the taxonomy through a multi-stage, human-AI collaborative process [24]. First, three authors iteratively created and consolidated an initial taxonomy over five sessions based on observations of online software task videos. Separately, we prompted *Gemini-2.5-Pro* to generate a taxonomy from scratch using our collected video dataset, without providing our initial version. We then augmented the human-generated taxonomy by integrating novel categories identified by the LLM. Finally, the combined taxonomy was validated against the entire video dataset to ensure comprehensive coverage and reorganized into the final set of nine distinct states. Our taxonomy aligns with Norman’s Seven Stages of Action [34], mapping Planning, Execution, and Evaluation to goal formation, action, and outcome assessment, and draws on Bloom’s cognitive hierarchy [4] that captures the shifts between operational

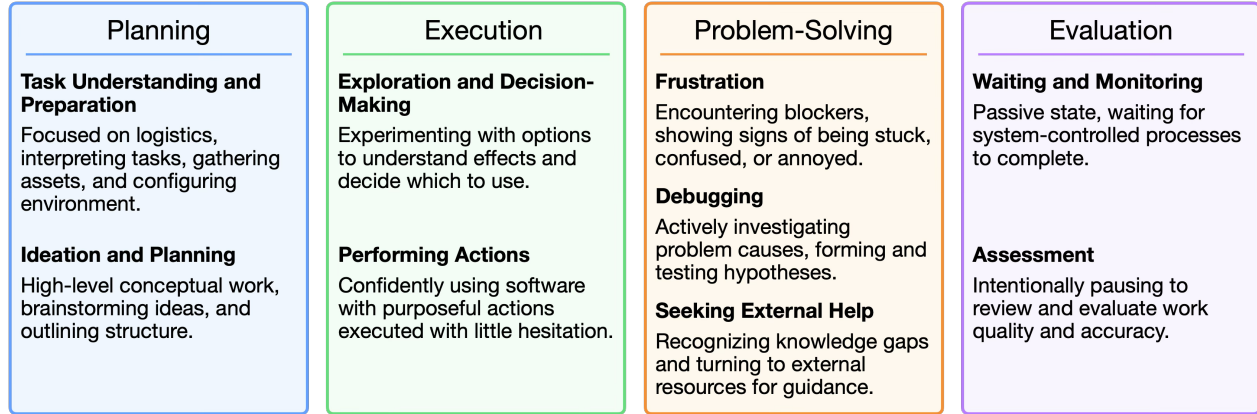


Figure 3. Our proposed taxonomy of user behavior states in GUI-based software tasks, organized into four main phases: **Planning**, **Execution**, **Problem-Solving**, and **Evaluation**. Each phase captures distinct patterns of user cognition and interaction, from initial goal formulation to iterative action, troubleshooting, and reflection.

(Execution) and critical work (Evaluation).

Dataset Curation. After constructing the taxonomy, we aligned each video with its corresponding narration segments. For every segment, we annotated the user’s behavior state using *Gemini-2.5-Pro* according to the taxonomy, prompting the model to produce both a predicted label and its reasoning. Two human annotators recruited from Prolific then verified and refined these annotations, achieving a 96.1% agreement rate. Finally, we uniformly sampled 200 instances from each of the nine classes, resulting in a balanced dataset of 1.8K annotated segments.

3.2.2. Intent Prediction

Description. This task evaluates whether a model can reason about the user’s short-term, immediate goal in context. It focuses on identifying what the user aims to achieve within open-ended workflows.

Dataset Curation. Using the narration-aligned video segments, we prompted *Gemini-2.5-Pro* to infer users’ intention in each segment. The think-aloud narrations often revealed users’ goals (e.g., “I’m going to align these objects”, “I’ll try another color”). Leveraging this signal, we prompted the model to infer the underlying user intention. After collecting and deduplicating the inferred intents, we further instructed the model to generate three plausible but incorrect alternatives to serve as distractors for the multiple-choice evaluation. The resulting intent annotations and distractors were then validated by the authors, with 88.68% of the data retained, yielding a final set of 1.3K instances.

3.2.3. Help Prediction

Description. The final task evaluates whether a model can progress from understanding and reasoning to deciding how to assist. Help Prediction consists of two subtasks: (1) **Help Need Detection**, a binary classification task that determines whether the user needs help, and (2) **Help Content Prediction**, which identifies the specific type of help needed, such as explaining a feature or suggesting an alternative.

Together, these subtasks assess a model’s ability to anticipate user needs and recommend appropriate assistance, bridging the gap between perception and actionable support.

Dataset Curation. We identified potential help-seeking moments using two complementary signals. First, *explicit help-seeking* behaviors, such as switching to external resources (e.g., Google, YouTube, ChatGPT), indicated direct attempts to seek guidance. Second, *implicit help-seeking* cues were extracted from user narration, where they expressed uncertainty or confusion (e.g., “How do I align this?”, “I can’t find Layer Mask.”). Additionally, we included clear *no-help-needed* moments, where users demonstrated confidence through their narration. Using these signals, *Gemini-2.5-Pro* was prompted to generate initial annotations for help-need and help-content labels. After deduplication, the model was additionally prompted to generate three plausible but incorrect options for each instance for multiple-choice question evaluation. All annotations and distractors were then reviewed by the authors, resulting in 1K validated instances, with 78.89% of the original data retained. For 12.5% of the retained instances, the segment’s start or end time was adjusted to exclude explicit visual help signals (e.g., user turning to Google Search) to ensure fair evaluation. Overall, 66% of the instances were labeled as help-needed, while the remaining 34% required no help.

4. Experiments

4.1. Experimental Setup

We evaluate a range of multimodal large language models (MLLMs) on our benchmark to assess their ability to understand, reason about, and assist users in open-ended software workflows. Our evaluation includes eight represen-

Model	(1) Behavior Detection		(2) Intent Prediction		(3) Help Prediction					
					Help Need Detection			Help Content Prediction		
	–	+ Prev.	–	+ Behavior						
					–	+ Behv.	+ Behv.+Intent	–	+ Behv.	+ Behv.+Intent
Gemini-2.5-Flash [15]	36.91	38.19	65.40	66.77	53.64	76.33	78.07	49.53	53.75	78.59
Gemini-2.5-Pro [15]	42.44	43.79	67.80	70.16	69.82	84.73	82.38	52.74	57.03	79.69
GPT-4o-mini [35]	17.65	17.07	60.76	62.19	46.05	78.92	82.26	31.32	42.86	79.84
GPT-4o [35]	36.32	37.24	61.19	62.58	49.69	87.79	87.91	45.95	48.37	79.78
Claude-4.5-Sonnet [1]	44.61	45.63	71.39	72.62	39.49	58.56	59.43	55.00	62.17	82.79
Qwen3-VL-8B [42]	37.97	38.13	62.70	64.03	52.83	70.39	77.36	46.06	50.63	80.11
InternVideo2.5-8B [43]	21.57	27.02	43.79	45.13	34.36	35.35	35.25	23.67	29.15	73.86
InternVL3-8B [57]	22.57	24.90	46.11	46.97	34.94	43.73	46.82	27.03	32.20	72.97

Table 2. Evaluation results on accuracy across (1) Behavior State Detection, (2) Intent Prediction, and (3) Help Prediction.

tative MLLMs spanning both proprietary and open-source models: **Gemini-2.5-Flash** [15], **Gemini-2.5-Pro** [15], **GPT-4o-mini** [35], **GPT-4o** [35], **Claude-4.5-Sonnet** [1], **Qwen3-VL-8B** [42], **InternVideo2.5-Chat-8B** [43], and **InternVL3-8B** [57]. All models are evaluated in a zero-shot setting using publicly available APIs or checkpoints, without any additional fine-tuning.

For each test instance, we uniformly sample 32 frames from the corresponding video segment, providing only visual input (excluding narration audio) to simulate perception based solely on visual cues. To ensure consistency across models, we use standardized prompting templates (Section H). We also prompt models to generate both a predicted label and supporting reasoning, a strategy shown to improve task performance [23].

Our main experiments are conducted in an offline inference setting, where models solve the task given the full video. To approximate real-world proactive assistant scenarios, we additionally evaluate an online setting, where the model receives visual input progressively—at 25%, 50%, 75%, and 100% of the segment, we uniformly sample 32 frames from the corresponding prefix for inference.

4.2. Evaluation Tasks

(1) Behavior State Detection. This task measures whether a model can identify the user’s behavioral state from a given video segment. We provide each model with clips and ask it to classify them into one of nine taxonomy-defined states. Two configurations are tested: (i) using only the current segment and (ii) with prior history, where the model is given the immediately preceding segment’s behavior state. This is framed as a multi-class classification problem, and performance is evaluated using accuracy.

(2) Intent Prediction. This task evaluates a model’s ability to infer the user’s underlying goal within a given video segment. Models are prompted to predict what the user’s

goal in two settings: (i) using only the current segment, and (ii) with additional behavior state context, where the model is also given the state label and its definition. We adopt a multiple-choice question (MCQ) format, where the model selects the most likely intent from four candidates. Performance is measured using accuracy. For the default setting (i), to mitigate potential bias, we additionally report multi-binary accuracy (MBAcc) following prior work [5, 7, 8], which evaluates whether the model correctly identifies the ground-truth intent in all three pairwise comparisons against incorrect alternatives.

(3) Help Prediction. The final task evaluates whether models can move beyond understanding and reasoning to provide actionable assistance. Given a video segment, models are asked to predict whether the user requires help (*Need*), and if so, what kind of help would be most appropriate (*Content*). **Help Need Detection** is framed as a binary classification task and evaluated using accuracy, precision, recall, and F1-score. **Help Content Prediction**, similar to Intent Prediction, uses a multiple-choice question (MCQ) format and is evaluated using accuracy and multi-binary accuracy (MBAcc) for the default setting. We test three settings for both tasks: (i) video only, (ii) video + behavior state, where the model is given the behavior label and its definition for the current segment, and (iii) video + behavior state + intent, where the model additionally receives the identified user intention. These settings progressively assess the model’s ability to leverage layered user context for meaningful, situation-aware assistance.

4.3. Results

Table 2 presents the performance of baseline models on GUIDE across the tasks, with accuracies reported under default and context-augmented settings. Overall, models performed weakest on Behavior State Detection and Help Prediction, with default-setting accuracies peaking at 44.61%

Model	Help Need Detection											
	–				+ Behavior State				+ Behavior State + Intent			
	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1	Acc	Prec	Rec	F1
Gemini-2.5-Flash [15]	53.64	83.27	36.62	50.87	76.33	97.67	65.47	78.39	78.07	94.56	70.62	80.86
Gemini-2.5-Pro [15]	69.82	76.42	78.09	77.42	84.73	93.61	82.34	87.61	82.38	91.20	80.94	86.76
GPT-4o-mini [35]	46.05	83.03	22.31	35.17	76.73	97.61	66.23	78.92	79.71	97.20	71.29	82.26
GPT-4o [35]	49.69	74.41	35.14	47.73	87.79	95.39	85.53	90.19	87.91	95.12	85.95	90.30
Claude-4.5-Sonnet [1]	39.49	87.69	8.92	16.19	58.56	99.16	37.09	53.99	59.43	99.19	38.44	55.41
Qwen3-VL-8B [42]	52.83	79.86	34.23	47.92	70.39	94.35	58.50	72.22	77.36	95.38	67.56	79.09
InternVideo2.5-8B [43]	34.36	33.33	0.16	0.31	35.35	90.91	1.56	3.07	35.25	83.33	1.56	3.07
InternVL3-8B [57]	34.94	72.73	1.25	2.46	43.73	98.88	15.77	27.20	46.82	98.40	19.22	32.16

Table 3. Results for **Help Need Detection** on accuracy, precision, recall, and F1-score across three conditions (default, with behavior state, with behavior state and intent).

and 55.00% for Behavior State Detection and Help Content Prediction, respectively, both from Claude-4.5-Sonnet [1]. While Gemini-2.5-Pro [15] reached nearly 70% accuracy on Help Need Detection, most other models showed substantially lower performance across both Help sub-tasks. Across tasks, we observe that models generally benefit from added behavioral and intent context, with particularly notable improvements in help-related predictions. We report the main findings below.

4.3.1. Behavior State Detection

Behavior State Detection remains highly challenging.

All models struggled to accurately infer the user’s behavioral state from video segments, underscoring the difficulty of the 9-way classification task. While proprietary models such as Claude-4.5-Sonnet [1] and Gemini-2.5-Pro [15] performed best, no model surpassed 45% accuracy, and most fell below 40%.

Models often misinterpret signals of struggle. The most common failure was misclassifying *Frustration* or *Debugging* as *Performing Actions* or *Exploration and Decision-Making*, as shown in the confusion matrix (Figure C4 in supp.). These errors suggest that models overlook subtle indicators of user difficulty—e.g., repeated clicks, hesitation, or undoing actions—instead interpreting them as deliberate progress, revealing a lack of nuanced understanding of user frustration signals.

Temporal context shows modest potential. Incorporating the prior behavior state led to small but consistent gains across models. While most improvements were marginal, the largest gain was observed for InternVideo2.5-8B [43] with 5.45 percentage points, suggesting that temporal context holds value and may be more effectively utilized with

improved temporal reasoning capabilities.

Model	Intent Prediction		Help Prediction	
	Acc	MBAcc	Acc	MBAcc
Gemini-2.5-Flash [15]	65.40	59.09	49.53	44.69
Gemini-2.5-Pro [15]	67.80	64.34	52.74	45.31
GPT-4o-mini [35]	60.76	50.24	31.32	28.59
GPT-4o [35]	61.19	56.58	45.95	41.25
Claude-4.5-Sonnet [1]	71.39	65.44	55.00	50.78
Qwen3-VL-8B [42]	62.70	58.07	46.06	44.69
InternVideo2.5-8B [43]	43.79	27.98	23.67	18.75
InternVL3-8B [57]	46.11	40.75	27.03	23.75

Table 4. Evaluation of **Intent Prediction** and **Help Content Prediction**, with Accuracy (Acc) and Multi-Binary Accuracy (MBAcc).

4.3.2. Intent Prediction

Intent Prediction is the most tractable task, but still imperfect. Among the three tasks, models achieved the highest performance on intent prediction, with several surpassing 60% accuracy. However, performance drops under the stricter MBAcc metric, which requires consistent discrimination across all answer pairs. This indicates that while models can often select a plausible intent, they still struggle with reliably identifying the correct one over all distractors (Table 4).

Behavior context helps, but only slightly. Incorporating behavior state context (i.e., the user’s behavioral label and definition) consistently improved performance, but the gains were relatively modest across all models. This suggests that while such context may offer useful cues, it does

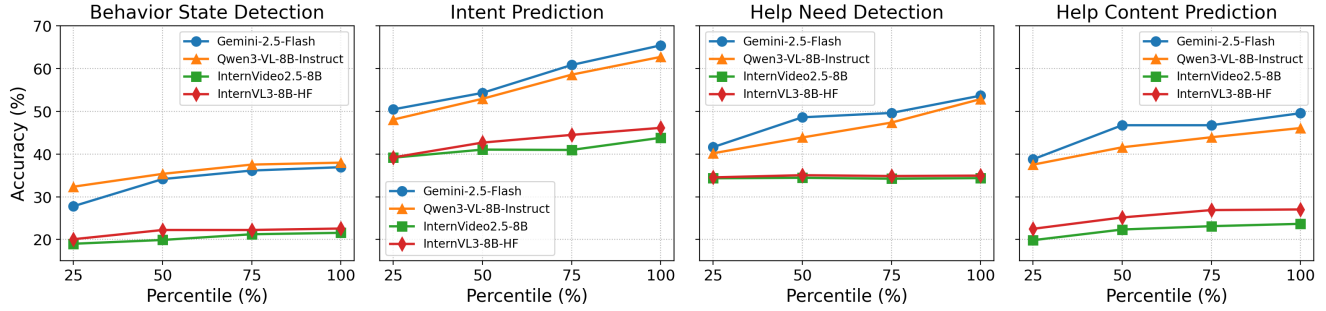


Figure 4. Accuracy trends across the tasks in the online setting, where models are given progressively more of the video segment (25%, 50%, 75%, and 100%). Models show consistent improvement as they see more segments, with Gemini-2.5-Flash [15] and Qwen3-VL-8B [42] showing larger and more consistent gains across all four tasks compared to the smaller open-source models.

not provide sufficient information on its own or is not yet effectively leveraged by current models for intent inference.

4.3.3. Help Prediction

High variance and missed help cases in Need Detection. Table 3 shows the full performance results for Help Need Detection. This subtask exhibited the most variance across models, with F1 scores ranging from 0.31 (InternVideo2.5-8B [43]) to 77.42 (Gemini-2.5-Pro [15]). Notably, recall was particularly low across most models—except for Gemini-2.5-Pro, all others had recall under 37%. This indicates that many instances where users actually needed help were misclassified as not needing it, echoing similar trends in Behavior State Detection (Section 4.3.1) where models frequently misinterpreted signals of struggle.

Behavior context improves Help Need Detection. Providing the user’s behavior state led to consistent and significant improvements in Help Need Detection across all models, with the largest gain observed in GPT-4o [35], with a 42.46-point increase in F1 score. This suggests that context, such as whether a user is exploring or showing signs of frustration, provides strong cues for determining help needs.

Help Content Prediction remains challenging, but benefits from intent context. Help Content Prediction proved particularly challenging, with all models struggling and the top accuracy reaching only 55% from Claude-4.5-Sonnet [1], which further declined to around 50% under the stricter MBAcc evaluation. However, incorporating user intent led to substantial improvements across models, with the largest gain in InternVideo2.5-8B [43] at 50.19 percentage points, highlighting the importance of understanding both user state and intent for providing targeted support.

4.3.4. Other Findings

Online vs. Offline Setting: models benefit more from temporal context. In our online simulation experiment,

where models are given progressively more of the video segment (25%, 50%, 75%, and 100%), we observe consistent performance gains across all four tasks (Figure 4). Gemini-2.5-Flash [15] and Qwen3-VL-8B [42] show larger and more consistent gains across all tasks, compared to the smaller open-source models, indicating a strong ability to integrate growing context into more accurate predictions. These findings suggest that gathering appropriate context over time is crucial for proactive AI assistance, where systems must not only react but also anticipate user needs based on incomplete and evolving information.

Outlook for Model Improvements. Together, these results suggest that incorporating structured user context (behavior state and intent) and temporal context consistently improves help prediction. Recent work on agents demonstrates the effectiveness of context engineering via stratified memory, where interaction history is selectively structured rather than treated as a flat sequence [49]. Applying this idea to GUI assistance is a promising direction for better leveraging long-horizon user context.

5. Conclusion

We introduced a benchmark for evaluating models in understanding, reasoning about, and assisting users in open-ended GUI-based workflows. Grounded in real-world novice user demonstrations, our tasks—behavior state detection, intent prediction, and help prediction—capture core capabilities needed for collaborative GUI agents. Evaluation across state-of-the-art MLLMs revealed that models struggle to interpret nuanced user behavior and accurately infer assistance needed in open-ended scenarios. However, when provided with appropriate user context, models showed consistent improvements, highlighting the value of structured user understanding. Overall, our benchmark provides a foundation for user-aware agents that support human workflows.

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